Computational Complexity, Homework Problem 3, Fall 2024

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Problem 3: *On input, we are given nonnegative integers d and k and two sets $A, B \subseteq \{0,1\}^d$, each of size n. We ask if there exist $a \in A$ and $b \in B$ such that the Hamming distance between a and b is at most k.*

Prove that if there exists a constant $\delta > 0$ and an algorithm solving the problem above in time $O(d^{100} \cdot n^{2-\delta})$, then the Strong Exponential Time Hypothesis fails.

Solution:

We will show that the orthogonal vectors problem can be reduced to this one. Given the sets A, B from the orthogonal vectors problem statement, we state that $O(A,B) = H(enc1(A), enc2(B), d)$, where $enc1$ and $enc2$ are encoding functions that will be defined later, O is the orthogonal vectors problem and H is the Hamming distance problem.

Define the encoding functions to work coordinate-wise on each vector on the set, expanding one bit into three bits as follows:

$$enc1(0) = (0,1,1),\ enc1(1) = (1,0,1),\ enc2(0) = (0,0,1),\ enc2(1) = (0,1,0)$$

Now, the distance between 0 and 0 is 1, 0 and 1 is 1 as well, while distance between 1 and 1 is 3. Therefore, distance between the full encodings is d iff distance between each encoded coordinate is 1, meaning the dot product is 0. Since if O can be solved in polynomial time on d, the SETH fails, we can't solve H in $O(d^{100} \cdot n^{2-\delta})$ time.